

Analysis of Household Surveys

A six-session workshop

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1 Introduction

Department of Economics, University of Ghana, hosted by the International Growth Centre (IGC) and the CEPR programme on Structural Transformation and Economic Growth (STEG). Five meetings of two hours, 11am–1pm, in CCC, and a sixth session online.

2 Administrivia

2.1 Meetings

Session	Day	Date	Topic
1	Tuesday	8 September	Design, sampling, and collection
2	Thursday	10 September	Describing welfare
3	Monday	14 September	Elicitation of consumption and survey design
4	Tuesday	15 September	Demand and welfare
5	Thursday	17 September	From cross-sections to dynamics
6	–	online	Capstone and frontiers

2.2 Texts

Angus Deaton’s *The Analysis of Household Surveys: A Microeconomic Approach to Development Policy* (Johns Hopkins, 1997) is the spine of the workshop. The World Bank’s 2018 reissue is open access, so it costs you nothing; the workshop page carries the link. Chapter 1 repays reading before the first session.

Two supplements, each for a particular session:

- Deaton & Zaidi, *Guidelines for Constructing Consumption Aggregates for Welfare Analysis* (World Bank LSMS Working Paper 135, 2002). Session 2.
- Deaton (1985), "Panel data from time series of cross-sections," *Journal of Econometrics* 30:109–126. Session 5.

2.3 Software

The hands-on work runs on a shared JupyterHub at hhsurveys.ligonresearch.org. You will be given a username and set your own password at first login (be sure to jot this down!). Nothing *must* be installed to take part, but I encourage you to also install the relevant software on your own computer (this will be necessary to use these tools after the workshop). The workshop page has instructions.

The computation uses `LSMS_Library`, which gives one interface to the household surveys of some forty countries. Installing it on your own machine is worth doing. Do it at home: it is a 1.3 GB download, and twenty laptops fetching it at once over classroom wifi will simply stall.

2.4 What you should be able to do afterwards

Not "know about," but "do." By the end of the sixth session you should be able to take an unfamiliar LSMS-style survey and:

1. Read its documentation well enough to say what population it represents, how it was drawn, and which of its variables the design lets you use;
2. Construct a defensible consumption aggregate from its modules, deflate it across space and time, and put an equivalence scale on it;
3. Produce weighted, correctly-clustered estimates of poverty and inequality, with standard errors you are willing to defend;
4. Estimate Engel curves and a simple demand system, and say what they imply for the welfare consequences of a price change;
5. Build a pseudo-panel from repeated cross-sections, and know when to prefer it to a real panel and when not;
6. Hand the whole thing to a colleague as code that runs.

2.5 Assessment

There is no assessment: this is a workshop, not a course. The sixth session is given over to a small analysis of a survey of your own choosing, which you will present to the room in about five minutes. Come to session 5 with a candidate country in mind.

3 Sessions

3.1 Design, Sampling, and Collection

3.1.1 Reading

- Deaton ch. 1 — the whole chapter
- Deaton §1.4 on survey design and §2.2 on what to do about it
- Skim the GLSS7 *Report* methodology section

Deaton is open access (CC BY 3.0 IGO). On the hub it is already in your `reading/` folder; otherwise PDF.

The GLSS7 report is published by the Ghana Statistical Service and is not ours to redistribute: GSS catalogue 97.

3.1.2 Brief description of LSMS Surveys

A timeline of the surveys `lsms_library` knows about: one row per country, one bar per survey wave, so we can see overall temporal coverage — 37 countries, 113 waves, 1985 to the present.

3.2 Describing Welfare

3.2.1 Introduction

3.2.2 The Theoretical Object

3.3 Elicitation of Consumption and Survey Design

3.3.1 Practical Difficulties in Measuring Individual Items of Consumption

3.3.2 Intra-Household Allocation

3.3.3 Constructing Aggregate Welfare Measures

3.4 Demand and Welfare

Where household surveys earn their keep for policy: what happens to whom when a price moves. Also: estimated demand functions can help solve a variety of measurement issues.

3.4.1 Introduction

1. Reading Deaton is in your **reading/** folder on the hub, or PDF.
 - Deaton, chs. 4 (§4.2 on Engel curves) and 5 (all)
 - Deaton & Muellbauer (1980), "An Almost Ideal Demand System," *AER* 70:312–326
 - Lewbel (1991), "The rank of demand systems," *Econometrica* 59:711–730
 - Ligon, "Estimating Household Welfare from Disaggregate Expenditures" (CUDARE working paper, 2020) — the CFE system we estimate today. In your **reading/** folder on the hub, or eScholarship.

3.4.2 Engel's Laws and Curves

1. Ernst Engel may be the first to use household data in statistical analysis, and used household budget data from Belgium and Saxony to propound what is sometimes called "Engel's (first) Law":

"The greater the income, the smaller the relative percentage of outlay for subsistence."

2. Budget shares and Engel curves Let $s_{ij} = p_j q_{ij} / x_i$ be household i 's budget share on good j , x_i total expenditure. The Working-Leser form,

$$s_{ij} = \alpha_j + \beta_j \log x_i + \gamma'_j z_i + u_{ij},$$

fits food shares remarkably well across countries and decades.

- Homothetic utility implies $\beta_{\text{food}} = 0$.
 - $\beta_{\text{food}} < 0$ is Engel's law
 - z_i : household size and composition, region, season
 - Estimated by OLS, clustered on the EA
3. Engel's Second Law A sort of corollary to the first law, sometimes translated as

"The proportion of the outgo used for food, other things being equal, is the best measure of the material standard of living of a population."

As a welfare measure, this solves some problems that the consumption aggregate has. In particular:

- Is a dimensionless share, so don't need to worry at all about price level, etc.
- If household size/scale can be expressed using Adult Equivalents A_i^θ , then those cancel also.
- Appears to be naturally comparable across time or place.

3.4.3 Other Important Elements of Demand Estimation

3.4.4 Constant Frisch Elasticity Demands

1. The estimating equation Take logs of expenditure on good j . Writing $w_i \equiv -\log \lambda_i$,

$$\log x_{it}^j = a_t^j + \beta_j w_{it} + \gamma'_j d_{it} + \varepsilon_{it}^j.$$

This is a **one-factor model**. The $(it) \times j$ matrix of log expenditures, once good-time effects a_t^j and characteristics d_{it} are swept out, is rank one: loadings β , factor w .

- estimate by SVD of the residual matrix, with missing data

- d_{it} enters as good-specific *Barten scales*, not "independence of base" adult equivalents.
 - needs **no prices** and **no total expenditure** — only expenditures on a subset of goods
2. $w = -\log \lambda$ as a welfare measure The multiplier λ is the same marginal utility of expenditure we talked about in session 2, so it *falls* as a household gets better off. Hence $w = -\log \lambda$ is increasing in welfare.
- Comparable across households facing different prices — the prices are in a_t^j , not in w
 - Comparable across household compositions — composition is in $\gamma_j' d$
 - Derived from behaviour, not from an assumed equivalence scale or an assumed price index
 - w is (minus) the partial derivative of indirect utility with respect to total expenditure: a genuine welfare object. It answers the question: for a 1% increase in budget, by what percent does utility increase?

3.4.5 Price Changes

1. Welfare consequences of a price change How much money makes household i as well off at prices p^1 as it was at p^0 ? With an expenditure function $e(u, p)$,

$$CV_i = e(u_i^0, p^1) - e(u_i^0, p^0), \quad u_i^0 = v(x_i, p^0).$$

For a small change in the price of one good j , this is approximately

$$CV_i \approx q_{ij} \Delta p_j = s_{ij} x_i \frac{\Delta p_j}{p_j}.$$

- to first order you need only the *budget share* — no elasticities
- the distributional incidence follows from how s_{ij} varies with x_i , which is the Engel curve you already estimated
- the exact figure needs the demand system, and says how far the approximation can be trusted

3.4.6 Exercises

3.5 From Cross-Sections to Dynamics

Two ways to get change out of survey data: build a panel from cohorts, or use one that already exists.

3.5.1 Reading

Deaton is in your `reading/` folder on the hub, or PDF.

- Deaton (1985), "Panel data from time series of cross-sections"
- Deaton, ch. 2 §2.7 and ch. 6
- Abadie, Athey, Imbens & Wooldridge (2023), "When should you adjust standard errors for clustering?" *QJE* 138:1–35

3.6 Capstone and Frontiers

Where the methods break down, and then your own analysis.

3.6.1 Reading